

# Mark O'Dowd

## SENIOR FULL STACK CLOUD ENGINEER

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### SUMMARY

Senior Full Stack Cloud Engineer with over 7 years of experience building and architecting scalable, cloud-native applications. Proven track record of bridging product vision with high-performance production while reducing infrastructure overhead through automated, cost-optimized deployments. AWS AI Certified, specialising in leveraging GenAI models and cloud ML services to build secure, enterprise-grade intelligent features

### TECHNOLOGY STACK

**Languages:** TypeScript, Python, Rust  
**Backend:** Django, Node.js, Express, Axum  
**Infrastructure:** AWS, GitHub Actions, CI/CD

**Frontend:** React, Next.js, Tailwind CSS  
**Databases:** PostgreSQL, SQLite, DynamoDB, MongoDB  
**Tools:** Linux, Bash, Docker, Neovim, Tmux

### CERTIFICATIONS

|                                   |         |                        |
|-----------------------------------|---------|------------------------|
| AWS Certified Solutions Architect | SAA-C03 | <a href="#">Verify</a> |
| AWS Certified AI Practitioner     | AIF-C01 | <a href="#">Verify</a> |
| AWS Certified Cloud Practitioner  | CLF-C02 | <a href="#">Verify</a> |

### EXPERIENCE

**Emerald Software** FULL STACK CLOUD ENGINEER Mar 2025 to Present

[www.fightorflight.studio](#) Developed a bespoke media focused portfolio website for Manchester based creative studio Fight or Flight leveraging Contentful for content management. Automated deployment pipelines for staging and production environments allowing the client to manage the platform. Implemented adaptive bitrate for video assets to improve performance and set email authenticity records (SPF/DKIM/DMARC) through Cloudflare

[www.irishtrad.ie](#) Architected a full stack web application featuring low latency interactive tools and educational modules for instruments used in traditional Irish music. The platform features a premium subscription and token payment flow processed with Stripe. The frontend uses server side rendering in Next.js and backend REST API developed in Rust (Axum). A variety AWS services were used in architecture such as Amplify, RDS, SES, S3

[www.jujitsu.ie](#) Using a modern tech stack I created a statically generated web application that digitised the World Ju-Jitsu Federation Ireland syllabus. An AWS S3 static hosting bucket is used allowing for an incredibly cost efficient application. Delivered a mobile first experience and optimised performance using WebP assets

[www.repair-rehab.com](#) Leading N.Ireland device repair shop Repair Rehab wanted to improve their online presence and operations online. A static landing page was created using React and a serverless architecture on AWS consisting of API Gateway, Lambda, DynamoDB used to create an order management platform

**Dawson Andrews** SOFTWARE ENGINEER May 2024 to Mar 2025

Dawson Andrews specialises in product design, strategy, and software development

- Rapidly adopted the Vue/Nuxt ecosystem to lead the end to end development of an enterprise grade web application ensuring a seamless transition from design to AWS backed production

**Hurree** SOFTWARE ENGINEER Jan 2023 to Mar 2024

Hurree is a cross platform analytics dashboard that enables teams to unify data to gain one high level overview

- Engineered user friendly analytics features using React and Django, unifying unique data sources into high-performance, real-time dashboards for cross-platform business intelligence while championing code quality and system reliability by leading technical design discussions and conducting rigorous code reviews

## Micro Focus SOFTWARE ENGINEER

Apr 2022 to Nov 2022

Micro Focus is one of the world's largest enterprise software providers delivering mission critical technology

- Spearheaded the integration of React into a mission critical legacy Java ecosystem creating a scalable i18n framework that enabled global localisation
- Provided mentorship and guidance to junior engineer on my team teaching how to do pull requests

## VML Technology

### WEB ENGINEER

Nov 2019 to Mar 2022

VML Technology pioneers new levels of video maximizing engagement through real time convergence of data

- Delivered core web platform features using React and Django, achieving 1:1 parity with mobile application for real time video engagement

## TECHNICAL LEADERSHIP & COMMUNITY

- Star Mentor:** [Zero to Mastery](#) is an online technology academy providing video courses on a wide range of topics with a Discord community of over 500,000 members. After 7+ years of dedication towards learning through the platform and giving back to the community through mentorship of juniors I am now 1 of 20 people in the community that has been recognised and granted the title of Star Mentor
- Tech Talks:** I am incredibly passionate about education and as a result I have spoken at tech events within the data engineering community covering topics such as data manipulation with Apache Spark and how to train your own model with Spark ML
- Meetups:** Engaging and networking within the various tech communities is important in order to stay on top of current trends and as such I have attended ~30 different events within the Dublin area: [Meetup Profile](#)
- Technical Blog:** I'm an avid book reader and have recently started doing a blog reviewing each book. The blog is available at [Personal Blog](#)
- Open Source Contributions:** As a Linux user I take the concept of open source to heart and try to give back to projects when I identify an issue or some way in which I can contribute. Some examples of contributions:
  - React Flow:** While working on a project that used the React Flow library I noticed issues during development with the TypeScript typings in the library itself so provided them a detailed explanation and discussion to fix: <https://github.com/xyflow/web/issues/479>
  - TanStack Table:** TanStack is a well regarded suite of libraries within the React ecosystem. When the Table product was early in development I was reading the docs and identified issues and provided a detailed Issue that included video which has since been completed: <https://github.com/TanStack/table/issues/5193>
  - Leptos Book:** Leptos is one of the leading frameworks in the Rust web ecosystem. While reading the documentation there was grammatical issues. I provided a fix: <https://github.com/leptos-rs/book/pull/223>

## EDUCATION

### Queens University Belfast

### BACHELOR OF SCIENCE IN MUSIC TECHNOLOGY

2012 to 2014